

introduce new legal complexities. Managing these risks requires not only technical solutions but also robust governance, regulatory alignment, and continuous monitoring. In the SDV era, trust becomes as critical as innovation.

In practical terms, the attack surface of an SDV extends across multiple layers, including in-vehicle networks, wireless interfaces such as Bluetooth and cellular connectivity, cloud APIs, and OTA update mechanisms. Securing this ecosystem requires a multi-layered approach. Encryption protocols are used to protect data both in transit and at rest, while secure boot mechanisms ensure that only authenticated software can run on vehicle hardware.

Hardware security modules (HSMs) embedded within automotive-grade processors provide cryptographic key management and secure execution environments, forming a root of trust within the system. Additionally, intrusion detection systems and continuous monitoring are increasingly deployed to identify anomalies in real time. As vehicles become connected computing platforms, cybersecurity is no longer optional; it is a foundational requirement for safety and trust.

India at the crossroads

India stands at a pivotal moment in the evolution of SDVs. The country possesses significant strengths, including a vast pool of software engineering talent and a rapidly growing digital ecosystem. At the same time, challenges remain, particularly in areas such as semiconductor manufacturing and

system-level automotive architecture. Bridging these gaps will be essential for India to fully capitalise on the opportunities presented by SDVs. Automakers like Tata Motors and Mahindra & Mahindra are already investing in this space, signalling the beginning of a broader transformation. The coming decade will determine whether India emerges as a global leader in SDV innovation or primarily remains a consumer market.

India's opportunity in the SDV space extends beyond manufacturing to include software, systems integration, and innovation ecosystems. A growing number of startups and engineering firms are working on vehicle software platforms, embedded systems, and AI-driven mobility solutions, complementing the efforts of established players like Tata Motors and Mahindra & Mahindra.

Government initiatives such as Make in India and production-linked incentive (PLI) schemes for electronics and semiconductors aim to strengthen domestic capabilities in critical components. However, bridging gaps in semiconductor fabrication, automotive-grade hardware design, and system-level architecture remains essential. By leveraging its strong software talent base and aligning policy with industry needs, India has the potential to become not just a consumer but a global hub for SDV development.

The trajectory of SDVs points towards a future defined by centralised computing, AI-driven intelligence, and cloud-native development. Vehicles will increasingly rely on digital twins for simulation,

enabling continuous improvement and faster innovation cycles. Subscription-based models will become more prevalent, and autonomy will gradually evolve as AI systems mature. Perhaps, most significantly, vehicles will no longer follow the traditional lifecycle of gradual decline. Instead, they will improve over time, gaining new capabilities and becoming more refined with each update.

At its core, the rise of SDVs represents a fundamental shift in how automobiles are perceived. A car is no longer a static product completed at the point of sale. It is a platform that evolves continuously through software. This transformation challenges long-held assumptions about ownership, value, and innovation. The idea of a 'finished' vehicle is becoming obsolete, replaced by a model of perpetual development and enhancement.

Modern vehicles are no longer defined solely by metal and mechanics, but by sophisticated software systems capable of learning, adapting, and evolving with every journey. In many ways, the industry is only at the beginning of this transformation. The future of mobility will not be built by software alone but by the seamless integration of software intelligence with robust electronic engineering foundations. **EFY**

This article is authored by Akanksha Sondhi Gaur, Senior Technology Journalist at Electronics For You (EFY), with valuable industry insights from Arpit Chahande, CEO of Yellow Matrix Private Limited; Akshay Raj Mathur, Lead Product Manager at Stellantis; Hariharan Lakshminarayanan, Group Product Manager at Stellantis; and Sukhendu Deb Roy, Industry Consultant.

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou

The latest from the Open Source world is here.

OpenSourceForU.com

Join the community at facebook.com/opensourceforu

Follow us on Twitter @OpenSourceForU